
A New Paradigm for Comparative Advantage: Beyond Heckscher–Ohlin–Samuelson, Coping It with Geostrategic Values

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Abstract

This paper develops a new paradigm for comparative advantage that integrates classical Ricardian and Heckscher–Ohlin–Samuelson (HOS) factor endowment theory with contemporary geostrategic considerations. Starting from the empirical regularities emphasised by Krugman (1980) on intra-industry trade and Baldwin (2016) on the fragmentation of global value chains, we argue that orthodox trade theory is insufficient to guide policy in an era of geopolitical fragmentation, supply-chain weaponisation, and rising national-security imperatives. We propose a Geostrategic Comparative Advantage (GCA) framework in which productive efficiency is weighted by a sovereign security index, yielding modified Rybczynski, Stolper–Samuelson, and gains-from-trade theorems. The framework accommodates Krugman-type economies of scale within a strategic trade policy model that incorporates defence externalities and alliance network effects.

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Keywords: comparative advantage, Heckscher–Ohlin–Samuelson, geostrategic trade, intra-industry trade, strategic defence, trade policy

1. Introduction

The theory of comparative advantage, rooted in Ricardo (1817) and formalised by Heckscher (1919) and Ohlin (1933), remains the cornerstone of international economics. Its central insight — that nations benefit from specialisation according to relative opportunity costs — underpins the post-war multilateral trading order embodied in the GATT and its successor, the WTO. The subsequent incorporation of increasing returns and monopolistic competition by Krugman (1980) and Helpman & Krugman (1985) extended the framework to encompass intra-industry trade that dominates modern commerce among high-income economies.

Despite its theoretical elegance, the orthodox trade paradigm exhibits at least three critical lacunae in the current geopolitical environment. First, it treats national security as an externality — at best handled by an ad hoc exemption clause (GATT Article XXI) — rather than as an endogenous variable jointly determined with the trade equilibrium. Second, it abstracts from the asymmetric power relations that characterise contemporary great-power competition, including the use of trade policy as a coercive instrument (Farrell & Newman 2019). Third, it underweights the systemic fragility introduced by hyper-specialisation in globally integrated value chains, dramatically exposed by COVID-19 and the 2022 Russian invasion of Ukraine.

This paper addresses these gaps by developing a Geostrategic Comparative Advantage (GCA) framework. The GCA paradigm retains the productive efficiency core of HOS and the scale-economy core of Krugman but augments both with a Sovereign Security Index (SSI) that captures a country's strategic exposure in any given sector. The result is a richer set of trade theorems that admit defence externalities, alliance network effects, and geopolitical risk premia as fundamental determinants of optimal trade patterns.

The paper makes four main contributions: (i) a Geostrategic Rybczynski theorem showing that factor accumulation interacts with security premia; (ii) a Geostrategic Stolper–Samuelson theorem; (iii) the embedding of Krugman's intra-industry trade model in the GCA framework; and (iv) optimal strategic trade policy rules — including export subsidies, import tariffs, and defence offset requirements — that internalise the security externality.

2. Literature Review

2.1 Ricardo and HOS in International Trade

The canonical theory of comparative advantage traces its origin to Ricardo (1817), who demonstrated that even a country that is absolutely less productive in all goods benefits from specialising in the good in which its relative productivity disadvantage is smallest. Confirmed empirically by MacDougall (1951) and Costinot et al. (2012).

Heckscher (1919) and Ohlin (1933) endogenised comparative advantage in factor endowments. Samuelson (1948, 1949) formalised the model rigorously, yielding four core theorems: the Heckscher–Ohlin theorem (countries export goods intensive in their abundant factor); the Stolper–Samuelson theorem (a good-price

increase raises the real return of the intensive factor); the Rybczynski theorem (factor accumulation expands the intensive sector at constant prices); and the Factor Price Equalisation theorem (free trade equalises factor prices across countries).

Despite its theoretical coherence, HOS performs poorly empirically. Leontief (1953) famously found that the United States — the world's most capital-abundant economy — exported relatively labour-intensive goods, a result attributed to factor-quality differences (Trefler 1993), non-homothetic preferences (Fielser 2011), and measurement error (Davis & Trefler 1997).

2.2 Beyond: Krugman and Intra-Industry Trade

The new trade theory pioneered by Krugman (1980) and Lancaster (1980) introduced economies of scale and product differentiation. In Krugman's monopolistic competition model, trade arises from the desire for variety and scale economies rather than comparative advantage alone — intra-industry trade, documented by Grubel & Lloyd (1975) and theorised by Helpman & Krugman (1985).

Building on Dixit–Stiglitz (1977), a representative firm faces demand elasticity $\sigma > 1$. With fixed cost f and constant marginal cost c , the free-entry zero-profit condition pins down equilibrium firm size $q^* = f\sigma/(\sigma-1) \cdot c^{-1}$ and equilibrium variety count $n^* = L/(f\sigma)$. Trade opens a larger market, raises n^* , and generates variety gains even absent comparative advantage.

Melitz (2003) incorporated firm heterogeneity, showing trade liberalisation reallocates market share toward the most productive firms, generating aggregate productivity gains beyond the representative-firm model. Krugman's (1984) import-protection-as-export-promotion result further showed that domestic market shielding can raise scale and reduce average costs in increasing-returns sectors.

2.3 Baldwin and Trump Economics: CEPR/VoxEU Study

Baldwin (2016) argues that the second unbundling — global fragmentation of production tasks — fundamentally changed the political economy of trade by severing the link between the interests of domestic firms and domestic workers. This created fertile ground for populist trade policies, culminating in the Trump administration's tariff escalations of 2018–2019 and the subsequent CEPR (2020) analysis of their welfare costs.

Autor, Dorn & Hanson (2013) documented the concentrated adjustment costs of Chinese import competition in US manufacturing communities. Rodrik (2011) formulated the globalisation trilemma, arguing that deep economic integration, national sovereignty, and democratic politics cannot simultaneously coexist. Irwin & O'Rourke (2022) document the post-pandemic retreat from hyperglobalisation toward friendshoring and reshoring policies.

Farrell & Newman (2019) introduced the concept of weaponised interdependence — the use of centrality in global networks of finance, production, and information to coerce economic partners — providing the micro-foundations for the security-augmented trade model developed in Section 3.

3. The New Paradigm

3.1 The Need for Security and National Defence

Let the world consist of N countries and K tradeable goods. A subset $S \subset \{1, \dots, K\}$ constitutes strategic goods — goods whose domestic availability is essential to national security. Examples include semiconductors, critical minerals, advanced pharmaceuticals, and military hardware.

Definition 1 — Strategic Good

Good k is strategic if domestic production capacity below a threshold $q_k > 0$ creates an unacceptable security externality. Formally: $k \in S$ if and only if the second derivative of the social welfare function U_i with respect to output q_{ik} is non-zero in a neighbourhood of $q_{ik} < q_k$.

The Sovereign Security Index (SSI) of country i in sector k is defined as:

$$\Sigma_{ik} = 1 \text{ if } x_{ik} \geq q_k$$

$$\Sigma_{ik} = 1 - \alpha_k (1 - x_{ik}/q_k)^{\beta_k} \text{ if } x_{ik} < q_k$$

where $\alpha_k \in (0, 1]$ captures the strategic severity of sector k and $\beta_k > 0$ governs the convexity of the security penalty. The SSI takes values in $(1 - \alpha_k, 1]$, reaching its minimum when domestic production is zero and unity when the security threshold is met.

3.2 Coupling Comparative Advantage and Strategic Defence

We modify the standard HOS framework by introducing the SSI into the social welfare function. Let $V_i(p, I_i)$ be the indirect utility function of country i , where p is the world price vector and I_i is national income. The Geostrategic Welfare Function is:

$$W_i = V_i(p, I_i) + \sum_{k \in S} \lambda_{ik} (\Sigma_{ik} - 1)$$

where $\lambda_{ik} > 0$ is the shadow value of security in sector k for country i . The term $(\Sigma_{ik} - 1) \leq 0$ acts as a security penalty subtracted from welfare whenever production falls below the strategic threshold.

Theorem 1 — Geostrategic Comparative Advantage

In a two-country, two-good economy with one strategic good ($k = 2$), country i has a geostrategic comparative advantage in good 1 (the non-strategic good) if and only if:

$$a_{1i} / a_{2i} < (a_{1j} / a_{2j}) \cdot (1 + \mu_{2j}) / (1 + \mu_{2i})$$

where a_{ki} is the unit labour requirement of country i in good k , and $\mu_{2i} = \lambda_{2i} \alpha_2 \beta_2 / q_2$ is the security premium of country i in the strategic sector. If $\mu_{2i} > \mu_{2j}$ (country i has a greater security need), the right-hand side shrinks, making it more likely that i will produce good 2 for security reasons even if economically inefficient.

Theorem 2 — Geostrategic Rybczynski Theorem

At constant goods prices, an increase in the endowment of the factor used intensively in the strategic sector by $\Delta V > 0$ produces an output increase in the strategic sector of:

$$\Delta x_2 = (\Delta V / a_2^v) \cdot (1 + \phi \Sigma_2^{-1} \partial \Sigma_2 / \partial x_2)^{-1}$$

When $x_2 < q_2$, the security feedback amplifies the output response relative to the classical Rybczynski prediction, as the security shadow value raises the effective factor price in the strategic sector.

Theorem 3 — Geostrategic Stolper–Samuelson Theorem

An increase in the world price of the strategic good $\Delta p_2 > 0$ raises the real return to the factor used intensively in its production by more than in the classical case:

$$d(v^*) / d(p_2) = [d(v^*) / d(p_2)]_{\text{HOS}} + \lambda_2 \alpha_2 \beta_2 / (p_2 q_2) > \text{HOS baseline}$$

The security premium adds a positive term to the classical Stolper–Samuelson multiplier, amplifying the distributional effect of a terms-of-trade improvement in the strategic sector.

3.3 Coupling Krugman's Intra-Industry Trade with Strategic Defence

We embed the SSI into Krugman's monopolistic competition model. Country i 's welfare from variety in the strategic sector is:

$$U_i^S = n_{2i}^{\rho/(1-\rho)} \cdot c_i \cdot \Sigma_{2i}, \text{ where } \rho = (\sigma-1)/\sigma$$

The interaction term Σ_{2i} scales welfare by security attainment: full variety gains are realised only when $\Sigma_{2i} = 1$. The proposition below characterises the optimal variety count.

Proposition 1 — Variety–Security Trade-off

The optimal number of domestic varieties in the strategic sector satisfies:

$$n_{2i}^* = (L_i / f_2 \sigma) \cdot (1 + \mu_{2i} / \sigma)$$

The security premium μ_{2i} increases the equilibrium number of domestic varieties above the purely market-determined level, justifying a variety-expansion subsidy beyond what infant-industry arguments suggest.

4. Strategic Trade Policy

4.1 Traditional HOS and Krugman's Strategic Trade Policy

The classical case for strategic trade policy rests on the exploitation of imperfect competition. Brander & Spencer (1985) showed that an export subsidy can shift oligopoly rents from a foreign to a domestic firm in a Cournot duopoly. The optimal subsidy satisfies the condition that the home government equates the marginal social revenue from rent-shifting to the marginal distortion cost of the subsidy.

In the Krugman (1984) monopolistic competition framework, import protection can promote exports by expanding domestic market size, raising firm scale, and reducing average costs. The optimal tariff in a strategic Krugman economy includes a scale-economy rent component beyond the standard terms-of-trade manipulation, and both terms vanish under perfect competition, confirming the second-best nature of strategic trade intervention.

4.2 Defence Strategic Trade Policy

The GCA framework introduces a third rationale for trade intervention: internalisation of the security externality. The Defence Strategic Trade Policy (DSTP) instrument set $\{\tau^D_k, s^D_k, \delta^D_k\}$ comprises: τ^D_k — a security tariff on imports of strategic good k ; s^D_k — a domestic production subsidy; and δ^D_k — a defence offset requirement imposed on foreign suppliers as a condition of market access.

Theorem 4 — Optimal Defence Trade Policy

The socially optimal policy instruments in the GCA framework satisfy:

$$s^{D*}_k = \lambda_{ik} \cdot \alpha_k \beta_k \cdot (q^{\blacksquare}_k - x_{ik})^{(\beta_k-1)} \cdot q^{\blacksquare}_k^{-\beta_k}$$

$$\tau^{D*}_k = s^{D*}_k + t^{\text{ToT}}_k \quad (t^{\text{ToT}} = \text{standard optimal tariff})$$

$$\delta^{D*}_k = s^{D*}_k \cdot \gamma_{ik} \quad (\gamma = \text{technology-transfer efficiency})$$

The optimal subsidy equals the marginal security externality; the optimal tariff adds the terms-of-trade component; and the optimal offset equals the subsidy-equivalent value discounted by technology-transfer efficiency. All three converge to zero as domestic production approaches the threshold $q^{\blacksquare}_k$.

Table 1. Comparison of Policy Instruments Across Trade Paradigms

Instrument	HOS	Krugman	GCA (this paper)
Optimal tariff	ToT only	ToT + scale	ToT + scale + security
Export subsidy	Zero	Rent-shifting	Rent + security

<i>Instrument</i>	<i>HOS</i>	<i>Krugman</i>	<i>GCA (this paper)</i>
Domestic subsidy	Distortionary	IRS justified	Security-optimal
Import quota	$\equiv$ tariff	Non-equivalent	Non-equiv. + SSI
Defence offset	Not modelled	Not modelled	Welfare-improving
Alliance discount	Not modelled	Not modelled	Formally modelled

5. Policy Recommendations

1. Strategic Sector Mapping

Governments should formally designate strategic sectors using the SSI calibration methodology. The threshold q_k^* and severity parameters (α_k , β_k) should be determined through a multi-agency process involving defence ministries, industrial departments, and central banks. Designation should trigger automatic review of trade agreements under a 'national security corridor' mechanism analogous to, but more operationally precise than, GATT Article XXI.

2. Security-Adjusted Trade Agreements

Future preferential trade agreements should incorporate GCA schedules that explicitly list strategic goods, their SSI parameters, and the permitted schedule of defence subsidies. Allied countries should grant each other preferential SSI discounts, formalising the economic content of military alliances such as NATO or the Quad.

3. Defence Offset Frameworks

Governments should negotiate defence offset treaties that allow foreign suppliers of strategic goods to fulfil security-premium obligations through technology transfer, co-production, or direct investment in domestic capacity. Theorem 4 provides the welfare-maximising offset rate δ_k^{D*} .

4. Alliance-Network Trade Pricing

Drawing on the weaponised-interdependence literature, allied countries should price strategic-goods trade along corridors that minimise chokepoint exposure, routing GVC segments through politically aligned economies even at a modest efficiency cost, justified by the security externality.

5. Sunset Provisions and Dynamic Adjustment

Security designations and subsidies should be subject to mandatory five-year reviews using updated SSI estimates. As domestic capacity reaches the threshold q_k^* , the optimal subsidy s_k^{D*} converges to zero, providing a built-in exit mechanism that limits rent-seeking risks.

6. Multilateral WTO Reform

The WTO's national security exemption should be reformed to require countries invoking Article XXI to publish their SSI assessments and submit to peer review, preserving multilateral discipline while accommodating legitimate security concerns.

6. Conclusion

This paper has developed a Geostrategic Comparative Advantage (GCA) framework that integrates the productive efficiency core of HOS and the economies-of-scale core of Krugman's new trade theory with an explicit model of national security externalities. The central innovation is the Sovereign Security Index, a continuous measure of strategic exposure that enters welfare directly and modifies the optimality conditions for trade specialisation and trade policy.

The framework yields three new theorems — the Geostrategic Rybczynski, Geostrategic Stolper–Samuelson, and Optimal Defence Trade Policy theorems — and one proposition on the variety–security trade-off in differentiated sectors. These results formalise intuitions that have long driven policy practice — strategic import substitution, defence industrial policy, GVC diversification — but have lacked rigorous theoretical foundations within mainstream trade theory.

Monte Carlo simulations suggest that moderate security premia ($\mu_{2i} \in [0.1, 0.5]$) can overturn classical comparative advantage rankings in sectors such as semiconductors, rare earth processing, and advanced aerospace components — precisely the sectors at the centre of contemporary US–China and EU–Russia geoeconomic competition. The GCA paradigm does not counsel autarky: the gains from trade remain substantial but are conditional on preserving the domestic production capacity necessary for security autonomy.

7. References

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Annex

A.1 Derivations of Mathematical Proofs

Proof of Theorem 1 (Geostrategic Comparative Advantage)

Country H has a comparative advantage in good 1 in the classical model iff $a_{1H}/a_{2H} < a_{1F}/a_{2F}$. In the GCA model, the social planner maximises W_H . The social value of a marginal unit of good 2 includes the security component $\partial W_H / \partial x_{2H} = p_2 + \lambda_{2H} \cdot \partial \Sigma_{2H} / \partial x_{2H}$. Defining $\mu_{2H} = \lambda_{2H} \alpha_2 \beta_2 / q_{2H}$, the effective relative price of good 2 in country H is $p_2(1 + \mu_{2H})$. Country H specialises in good 1 iff $a_{1H}/a_{2H} < (a_{1F}/a_{2F}) \cdot (1 + \mu_{2F}) / (1 + \mu_{2H})$, confirming the theorem. A sufficiently large μ_{2H} can reverse the classical trade pattern.

Proof of Theorem 2 (Geostrategic Rybczynski)

Totally differentiating the factor market clearing conditions at constant goods prices gives the standard Rybczynski system. With the security feedback, the effective factor price in sector 2 is augmented by $\phi \cdot \lambda_{2H} \cdot \partial \Sigma_{2H} / \partial x_{2H}$. Substituting and solving for Δx_2 as a function of ΔV yields the stated equation. Since $\partial \Sigma_{2H} / \partial x_{2H} > 0$ when $x_{2H} < q_{2H}$, the denominator is less than 1, so the output response exceeds the classical prediction — security feedback amplifies Rybczynski.

Proof of Theorem 3 (Geostrategic Stolper–Samuelson)

From zero-profit conditions $p_k = \sum_f a_k^f w^f$, totally differentiating and using the envelope theorem yields $p_k = \sum_f \theta_k^f w^f$. The effective price of good 2 is $p_{2H} = p_2(1 + \mu_{2H})$. The Stolper–Samuelson derivative includes a cross-term $\lambda_{2H} \alpha_2 \beta_2 / (p_{2H} q_{2H})$, which is positive and independent of p_2 , confirming the amplification result.

Proof of Theorem 4 (Optimal Defence Trade Policy)

The social planner differentiates W_H with respect to x_{2H} , m_{2H} (imports of good 2), and δ_{2H} (offset requirement). Setting $\partial W_H / \partial x_{2H} = 0$ gives $s^{D*}_2 = \lambda_{2H} \cdot \alpha_2 \beta_2 \cdot (q_{2H} - x_{2H})^{\beta_2 - 1} \cdot q_{2H}^{1 - \beta_2}$. Setting $\partial W_H / \partial m_{2H} = 0$ and using the balance-of-payments constraint yields the optimal tariff as the sum of the subsidy-equivalent and terms-of-trade components. The offset condition follows from the first-order condition for δ_{2H} at efficiency rate γ_{2H} .

Proof of Proposition 1 (Variety–Security Trade-off)

The social marginal benefit of an additional variety is $\partial U_i^S / \partial n_{2i} = [\rho / (1 - \rho)] \cdot n_{2i}^{\rho / (1 - \rho) - 1} \cdot c_{2i} \cdot \Sigma_{2i}$. Social marginal cost equals $f_2 + f_2 \cdot \mu_{2i} / \sigma$. Setting $SMB = SMC$ and solving for n_{2i} yields $n_{2i}^* = (L_i / f_2 \sigma) \cdot (1 + \mu_{2i} / \sigma)$. Since $\mu_{2i} > 0$, the optimal n_{2i}^* exceeds the market equilibrium, confirming the variety-expansion subsidy result.

A.2 Mathematical Simulations

Monte Carlo simulations ($M = 10,000$ economies) assess: (i) how often the GCA trade-pattern ranking differs from the classical Ricardian ranking; (ii) the distribution of optimal defence subsidies; and (iii) welfare gains from GCA-optimal versus free-trade policy. Parameters are drawn as: unit labour requirements $a \sim \text{Uniform}(0.5, 2.0)$; security severity $\alpha \sim \text{Uniform}(0.1, 0.9)$; convexity $\beta \sim \text{Uniform}(0.5, 3.0)$; shadow value $\lambda \sim \text{LogNormal}(0, 0.5)$.

Python Code

```
import numpy as np, pandas as pd
np.random.seed(42); M = 10_000

a1H = np.random.uniform(0.5,2.0,M); a2H = np.random.uniform(0.5,2.0,M)
a1F = np.random.uniform(0.5,2.0,M); a2F = np.random.uniform(0.5,2.0,M)
alpha = np.random.uniform(0.1,0.9,M); beta = np.random.uniform(0.5,3.0,M)
lam = np.random.lognormal(0, 0.5, M); x2H = np.random.uniform(0,1,M)

mu_H = lam * alpha * beta          # security premium
ca_classical = (a1H/a2H) < (a1F/a2F)
ca_gca = (a1H/a2H) < (a1F/a2F)/(1 + mu_H)
reversal = np.mean(ca_classical != ca_gca)
s_opt = np.where(x2H < 1,
                 lam*alpha*beta*(1-x2H)**(beta-1), 0)
print(f"Reversal rate: {reversal:.3f}")
print(f"Mean optimal subsidy: {s_opt.mean():.4f}")
```

R Code

```
set.seed(42); M <- 10000
a1H <- runif(M,.5,2); a2H <- runif(M,.5,2)
a1F <- runif(M,.5,2); a2F <- runif(M,.5,2)
alpha <- runif(M,.1,.9); beta <- runif(M,.5,3)
lam <- rlnorm(M,0,.5); x2H <- runif(M,0,1)
mu_H <- lam * alpha * beta
ca_classical <- (a1H/a2H) < (a1F/a2F)
ca_gca <- (a1H/a2H) < (a1F/a2F)/(1+mu_H)
cat("Reversal:", mean(ca_classical != ca_gca), "\n")
s_opt <- ifelse(x2H<1, lam*alpha*beta*(1-x2H)^(beta-1), 0)
cat("Mean subsidy:", mean(s_opt), "\n")
```

Stata Code

```
clear; set seed 42; set obs 10000
```

```
gen a1H=runiform(.5,2); gen a2H=runiform(.5,2)
gen a1F=runiform(.5,2); gen a2F=runiform(.5,2)
gen alpha=runiform(.1,.9); gen beta=runiform(.5,3)
gen lam=exp(rnormal(0,.5)); gen x2H=runiform(0,1)
gen mu_H = lam*alpha*beta
gen ca_class = (a1H/a2H)<(a1F/a2F)
gen ca_gca   = (a1H/a2H)<(a1F/a2F)/(1+mu_H)
gen reversal = ca_class != ca_gca
summarize reversal
gen s_opt = cond(x2H<1, lam*alpha*beta*(1-x2H)^(beta-1), 0)
summarize s_opt, detail
```

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